Vitor Pinheiro

AI/ML Software Engineer
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SUMMARY

AI/ML engineer with 5 years building production ML and LLM systems in fintech. Shipped a LangGraph support agent that resolves 80% of internal tickets, an LLM explainability layer for fraud alerts, and a LightGBM fraud model that cut false-positive alerts by 99%. Comfortable across the stack, from Spark pipelines and Databricks MLOps to agents and evaluation.

SKILLS

- **Proficient:** Python, Scala, SQL, Spark/PySpark, Databricks, Streamlit, GitHub Actions, Azure
- **Experienced:** Docker, LangChain, LangGraph, MLflow, Keras, TensorFlow, OR-Tools, Claude, Cursor, LightGBM, XGBoost, SHAP, Delta Live Tables, Unity Catalog
- **Familiar:** FastAPI, Kubernetes, Airflow, Clojure, JavaScript/TypeScript, HTML, CSS
- **Domain:** credit/lending, fraud detection and transaction monitoring, time-series forecasting, LLM agents and evaluation, MLOps

EXPERIENCE

Qubika · AI/ML Engineer (client: Coastal Community Bank)
Remote, part-time · Feb 2026 – Jul 2026

- Productionized a LightGBM fraud detection model that, on a 24-month backtest, cut transaction-monitoring alert volume by 99.2% (12.9M → 98.5K) and lifted analyst-facing precision from 1.29% to 76.4%, a 131× reduction in review burden while still surfacing 45% of confirmed fraud.
- Built the MLOps platform for the fraud models with Delta Live Tables, Databricks Asset Bundles, MLflow and Unity Catalog.
- Built an explainability pipeline that turns each fraud alert into a reviewer-ready rationale by computing SHAP attributions and calling a Databricks-hosted foundation model through Mosaic AI Model Serving with structured prompts, producing customer- and transaction-context narratives that speed up case triage.

Nubank · AI/ML Software Engineer
São Paulo, BR (hybrid) · Sep 2022 – Jul 2026 · Intern → Jr Engineer → Engineer

- Built a Slack AI assistant using Python, LangChain, LangGraph and OpenAI over the framework's codebase, documentation and Slack channel history, deployed in production with Docker/Kubernetes, autonomously resolving 80% of support tickets (10 → 2/week) and unlocking engineering hours for infrastructure and feature improvements.
- Evaluated Databricks Genie as a self-service analytics assistant for the Lending Business Unit through a proof of concept with business users, handling 3 of 5 test scenarios and documenting maturity gaps that informed the adoption strategy.
- Translated the Lending NPV framework from Scala to Python/PySpark, enabling unified profitability analysis across real-time credit models and services with both distributed and single-node simulations of loan performance and decision economics.
- Generated cleaner, less biased training data for personal-loan models by co-designing Scala/Spark experimentation logic that randomized offer parameters on customer subsamples, reducing survivorship bias and improving future credit policy decisions.
- Reduced landing time for critical Lending core policy datasets by 34 hours by reordering ETL execution, pushing filters earlier in the Scala/Spark DAG and applying time-based partitioning, improving data freshness for real-time credit models.
- Turned a ~2-day manual reporting process into a weekly task of under an hour by co-developing an internal low-code dashboard tool in Python/Streamlit, driving adoption from Lending to 19+ business units and letting several teams replace Looker at ~1 USD/day in infra costs.
- Built analytics abstractions (filters, aggregations, re-aggregated monitorings) for an internal low-code dashboard framework, cutting execution time from ~20 minutes to ~5 seconds.
- Led data governance initiatives for Lending, including the subdomain migration, SCR monitoring infrastructure and credit metrics catalog consolidation, moving 100% of the team onto the new infrastructure.

EloGroup · Data Science and Engineering Intern (client: Comgás)
São Paulo, BR (remote) · Jun 2021 – Sep 2022

- Helped cut natural-gas contractual fines by roughly R$200M in the first three months by performing EDA, sanity checks and time-series modeling over 24 city-level series for Comgás, enabling an ensemble forecast that kept 18 of 24 citygates within the 5% fine threshold.
- Built Azure ML pipelines from Data Lake to Power BI (Data Factory, Databricks, Synapse) with MLflow model tracking and automated data quality checks in a medallion architecture.

PROJECTS

- **CarDeck** (github.com/jvitorcosta/spotify-car-deck): open-source ESP32 firmware that turns a dashboard screen into a Spotify now-playing display styled as a Pokémon battle, with synced lyrics, a companion screen and a Bluetooth OBD gauge cluster kept read-only by a tested allow-list (~320 commits in two weeks, CI and tagged releases).

- **SolvedOR** (github.com/jvitorcosta/or-solver-streamlit): a Streamlit app (my bachelor thesis) that teaches Operations Research: students write models in a custom syntax, OR-Tools solves them, and Plotly shows the feasible region in up to 3D.

EDUCATION

Bachelor of Industrial Engineering, Universidade Federal do Amazonas (UFAM), Manaus, BR · Jan 2019 – Sep 2023 · GPA 9.07/10

CERTIFICATES

- Google Data Analytics Professional Certificate, Coursera · June 2023
- Information Technology Technician, Matias Machline Foundation · December 2016

LANGUAGES

Portuguese (native) · English (advanced, C1) · Mandarin (intermediate, B1)